```
#include <Servo.h>

#define TRIG_PIN D5
#define ECHO_PIN D6
#define SERVO_PIN D7
#define BUZZER_PIN D8

Servo lidServo;

long duration;
int distance;
bool lidOpen = false;
unsigned long lidOpenTime = 0;
const unsigned long lidOpenDuration = 3000; // 3 seconds
const int openDistanceThreshold = 15; // Distance in cm to trigger lid open
const int binHeight = 30; // Height of the bin in cm
const int fullThreshold = 80; // Percentage threshold to consider bin full

void setup() {
  Serial.begin(9600);
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  lidServo.attach(SERVO_PIN);
  lidServo.write(0); // Ensure lid is closed
  digitalWrite(BUZZER_PIN, LOW);
}

void loop() {
  // Measure distance to detect hand
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
```

```
digitalWrite(TRIG_PIN, HIGH);
delayMicroseconds(10);
digitalWrite(TRIG_PIN, LOW);
duration = pulseIn(ECHO_PIN, HIGH);
distance = duration * 0.034 / 4;

Serial.print("Distance: ");
Serial.print(distance);
Serial.println(" cm");

// Open lid if hand is detected
if (distance < openDistanceThreshold && !lidOpen) {
  lidServo.write(180); // Open lid
  lidOpen = true;
  lidOpenTime = millis();
  Serial.println("Lid opened");
}

// Close lid after specified duration
if (lidOpen && (millis() - lidOpenTime >= lidOpenDuration)) {
  lidServo.write(0); // Close lid
  lidOpen = false;
  Serial.println("Lid closed");
}

// Measure bin fullness
int binDistance = measureBinDistance();
int fillLevel = map(binHeight - binDistance, 0, binHeight, 0, 100);
fillLevel = constrain(fillLevel, 0, 100);
Serial.print("Fill Level: ");
Serial.print(fillLevel);
Serial.println("%");
```

```
  // Trigger buzzer if bin is full
  if (fillLevel >= fullThreshold) {
    digitalWrite(BUZZER_PIN, HIGH);
    Serial.println("Bin is full! Buzzer activated.");
  } else {
    digitalWrite(BUZZER_PIN, LOW);
  }

  delay(500); // Short delay before next reading
}

int measureBinDistance() {
  // Function to measure distance from sensor to trash
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);
  long duration = pulseIn(ECHO_PIN, HIGH);
  int distance = duration * 0.034 / 2;
  return distance;
}
```